

A Model for the Molecular Basis of Filament Contractility and Sliding as Demonstrated by Helix Models

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The twisting behavior of α helices has hardly been considered hitherto with regard to the function of proteins. The well-known electrostatic repulsion between the highly charged side chains, which depends on their interaction with ions, is absolutely connected with torsional rotations of the helix as long as its hydrogen bonds hold. This means a direct transformation of chemical into mechanical energy. However, the stability of a twisted single α helix with charged side chains is low in an aqueous environment. It may easily ball up to form a globular molecule with nonhelical regions of the polypeptide chain. This corresponds to a primitive contraction that obviously occurs with spasminlike proteins that contain strongly twisted filaments as Salisbury [J. Submicrosc. Cytol. 15:105–110, 1983] has shown. Steps that increase the stability and rigidity of α helical filaments are 1) the formation of coiled-coils, 2) self-intertwining (“telephone cord phenomenon”) or intertwining with other coiled-coils as shown with the intermediate filaments, and 3) association with cytoskeletal elements (microfilaments, protofilaments of microtubules) that contain globular subunits. These coarser elements are rotated by winding and unwinding of the smaller helical molecules and thus transmit the torsion produced in the α helices to the microscopic level by the sliding (screwing) motion and the shearing effect that is connected with the waves of a rotating helix. Particles are transported if connected to the helical side arms. Since the displacement of the side arms seems to occur along the single protofilaments of a microtubule, a rotation of these protofilaments is suggested. The bidirectional transport of particles along single microtubules may be explained by the association of left- and right-handed helices with the protofilaments. According to the models, parallel and antiparallel sliding of neurofilaments and neurotubules is suggested.

Key words: α -helix, filament motility, filament contractility, filament sliding, microtubules

INTRODUCTION

A model for filament motility should agree with the known facts of protein structure and should consider the mechanical characteristics of protein filaments, eg, stiffness, elasticity, and capacity to take up torsional force. These are not comprehensible by the pure inductive methods of cell biology or chemistry. However, similar characteristics are shown by elastic helices, eg, those made of steel wire. This paper describes some model experiments with helices that may be very important for the mechanical behavior of protein filaments.

TORSIONAL ROTATION AND COLLAPSE OF A SINGLE α HELIX

Since the length of the hydrogen bonds in an α helix depends on the electrostatic repulsion between the highly charged side chains [eg, Poland and Scheraga, 1970], each discharge after ion addition may decrease this repulsion and result in a shortening of hydrogen

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bonds. As shown, eg, by the simplified α helix model of Fraser and MacRae [1973], a change of the unit height is absolutely connected with a change in the unit twist. This can be demonstrated in the best way by shortening or stretching of a long tubular helix model as shown in Figure 1: A pitch decrease is connected with torsional rotations (arrow r). This means a regular and directed nonthermodynamic motion. However, the stability of a charged α helix is small in an aqueous environment [Schellman, 1955]. Hydrogen bonds may easily break, and a globular region can be formed, as is demonstrated with the tubular α helix model that has been incised and then twisted (Fig. 2). A torsional force can only be generated as long as the hydrogen bonds hold. The globular domain would take up all that force. Its formation appears as a primitive contraction. Its site is sequence-determined. Some torsional force may be involved during the "induced fit" motion of enzymes. A simple model to simulate the behavior of a bundle of twisted α helices (or short coiled-coils) of low stability is a twisted rubber string with some weak points (Fig. 3a). This probably corresponds to the behavior of spasminlike filaments that produce Ca-controlled contraction in many protozoa [Salisbury, 1983; and this symposium]. Its irregular self-intertwining is shown in Fig. 3b. A presupposition for this primitive contraction is the lack of tension. A strongly stretched string cannot intertwine and contract. The weak tension directed to the middle (arrows T) is generated only after intertwining has started.

THE TORSIONAL ROTATION OF A COILED-COIL

A special amino acid sequence results in the very stable coiled-coil configuration. Here, two α helices are intertwined and connected by strong hydrophobic bonds [Mac Lachlan and Stewart, 1975]. The expected torsional conformation change after ion addition to a highly charged coiled-coil [Jarosch, 1978, 1985] is shown in Figure 4. The rubber string model of Figure 5 demonstrates the possible motion of a single coiled-coil as shown by spectrin, filamin, and similar filamentous protein filaments. The screwing motion of the free ends is directed here to the middle of the molecule (arrows S) and causes self-intertwining ("telephone cord phenomenon"). The newly formed double helix is right-handed and contains two antiparallel coiled-coils [Jarosch, 1985]. Self-intertwining can be prevented by increasing the stiffness of the twisted filament. One possibility is to form a stable rope by intertwining many coiled-coils or pairs of antiparallel coiled-coils. This is obviously the case with intermediate filaments, in which the destabilizing globular domains are eliminated by a staggered arrangement of the monomers [eg, Crewther et al, 1983]. The other possibility is to associate with the coarser cytoskeletal

elements (microfilaments, protofilaments of microtubules) and to rotate them by winding and unwinding along the whole length. It seems that this association is necessary to transfer the torsional motion effectively to the microscopic level and to cause the quick modes of cell motility.

SLIDING AND SHEARING

A rotating helix produces helical waves (arrows W in Fig. 6) that are indeed apparent waves that cause a shearing force in the surrounding medium, which screws (or slides) the helix into the direction opposite to the waves (arrow S). In the model of Figure 6, the helix is wrapped around a rod and is rotated by the winding of an electromotor-driven helix. A particle connected to that helix at the site of winding is transported (long arrow). The resistance of the environment determines which of the two motions (waves or sliding) dominates. The model of Figure 7 shows waves only (no sliding) and moves two particles. Note the length change of the numbered side arms. However, a single molecular helix, eg, a coiled-coil, cannot produce waves and particle transport because the resistance of the medium is too strong. It can screw and coil only. The same is true for a single microfilament. Vigorous cytoplasmic streaming, eg, in *Nitella* cells [Foissner and Jarosch, 1981], is shown only by a bundle of microfilaments. The velocity of the particle transport depends on the number of rotating helices [Jarosch, 1972].

THE TRANSPORT OF PARTICLES ALONG MICROTUBULES

The author believed for a long time that microtubules (MTs) can move and transport particles by rota-

Figs. 1-5. The mechanical behavior of single α helices and coiled-coils as demonstrated by models.

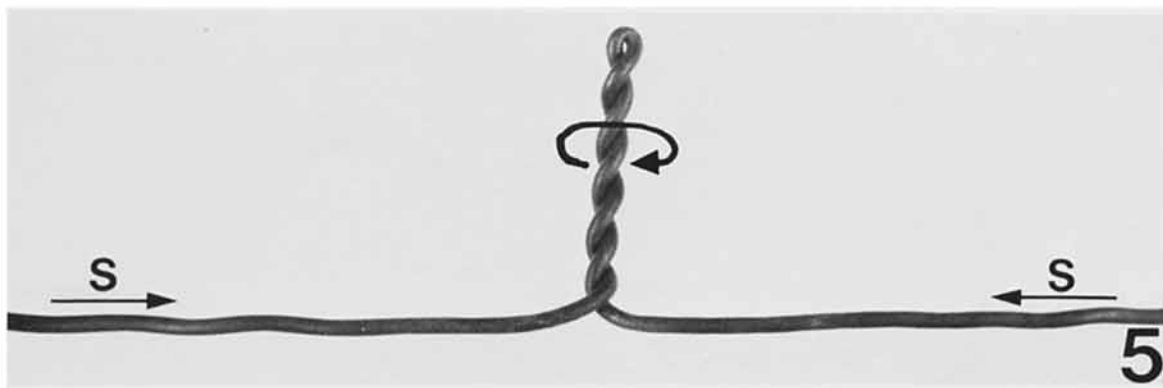
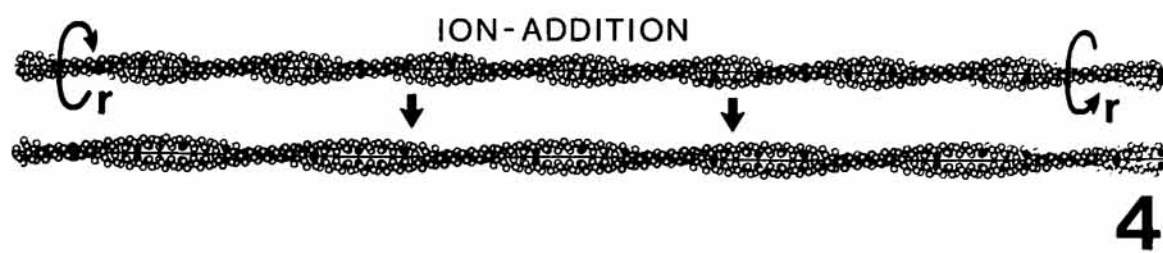
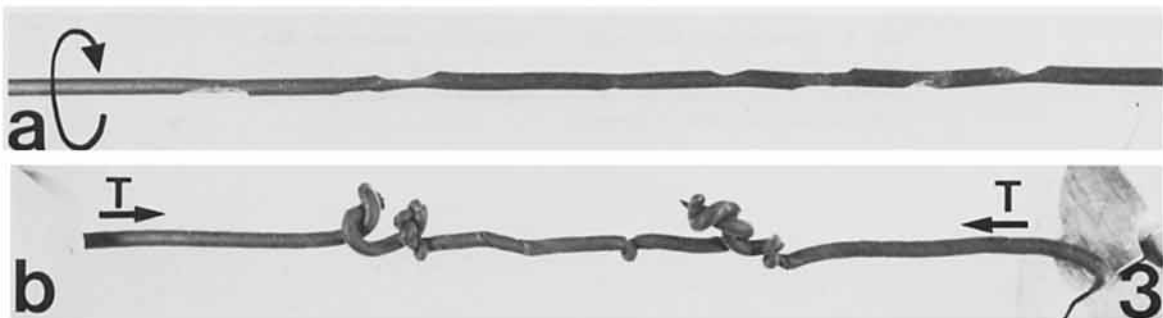
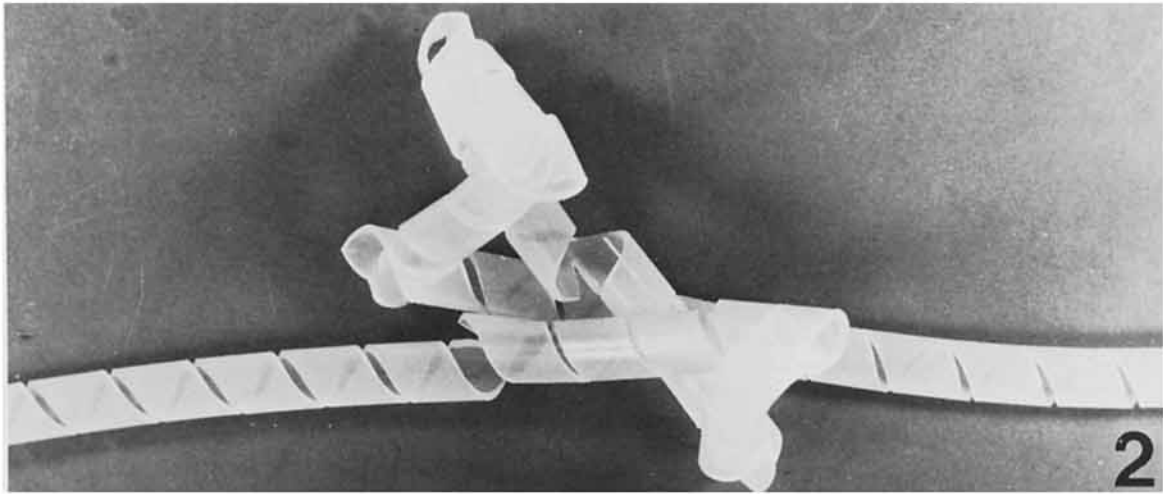
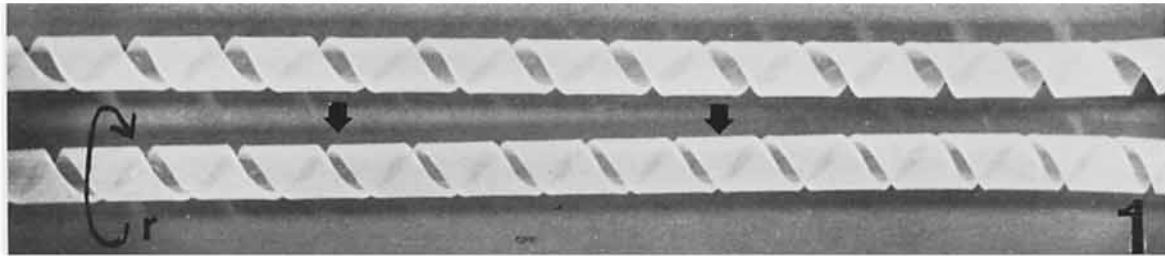
Fig. 1. A helical plastic tube represents a simplified model of an α helix. A pitch change as shown produces torsional rotations (arrow r).

Fig. 2. A globular transformation takes place with the region of the twisted helix model that has been incised previously.

Fig. 3. The collapse of a twisted rubber string with some incisions simulates the behavior of a spasminlike filament.

Fig. 4. The conformational change of a coiled-coil after ion addition produces strong torsional force by a little uncoiling of the two α helices (arrows r).

Fig. 5. The self-intertwining of a twisted rubber string ("telephone cord phenomenon") simulates the behavior of a single coiled-coil filament.

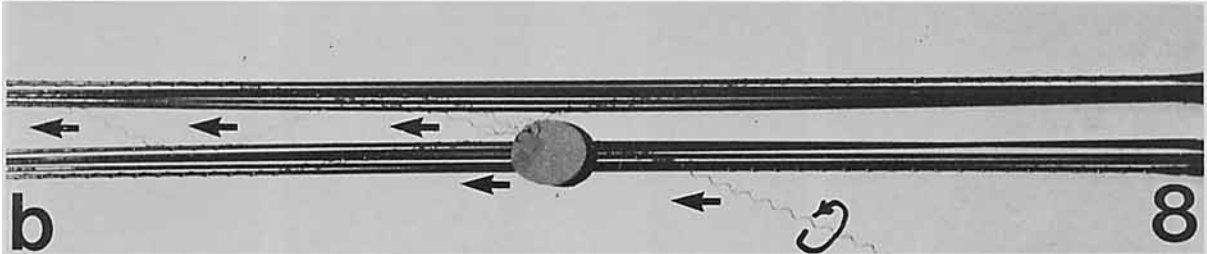
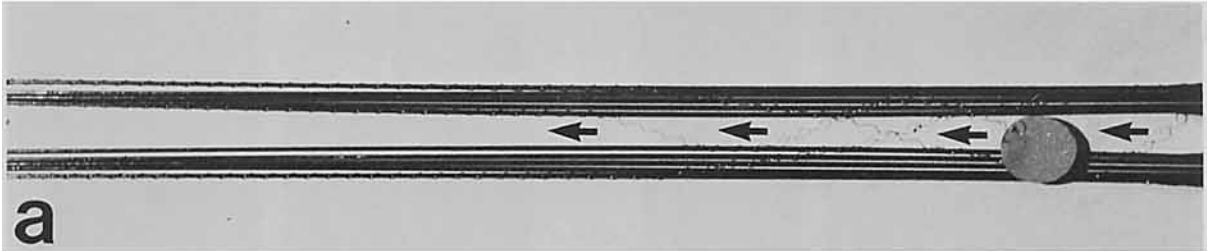
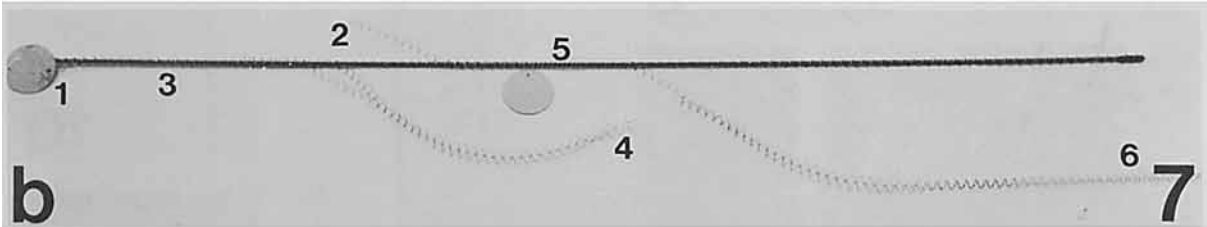
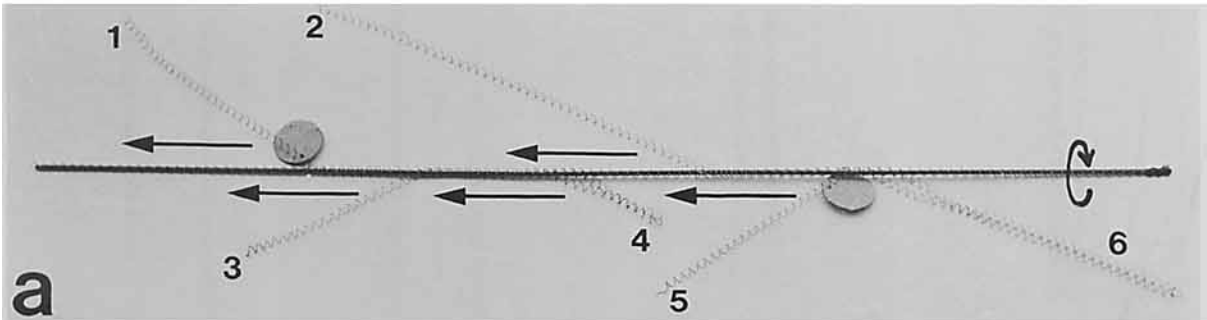
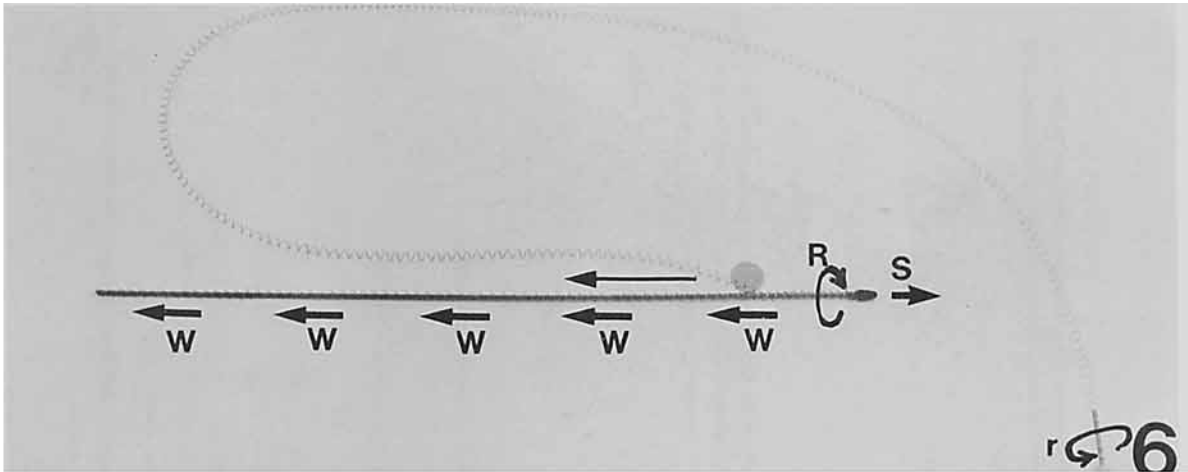


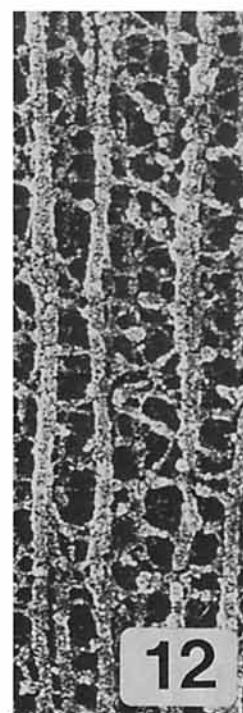
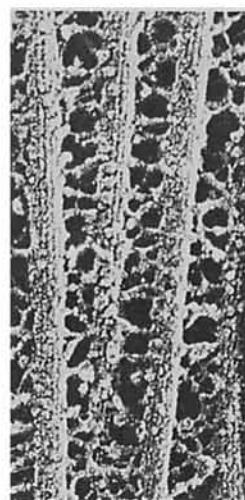
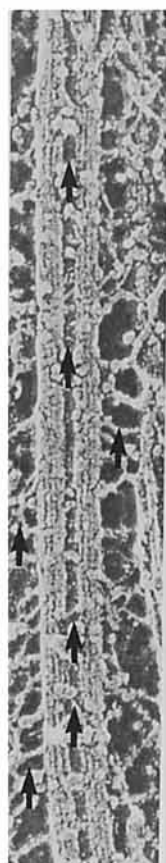
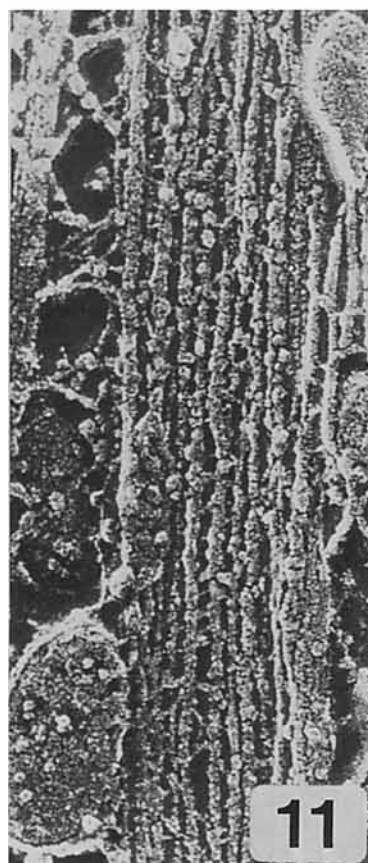
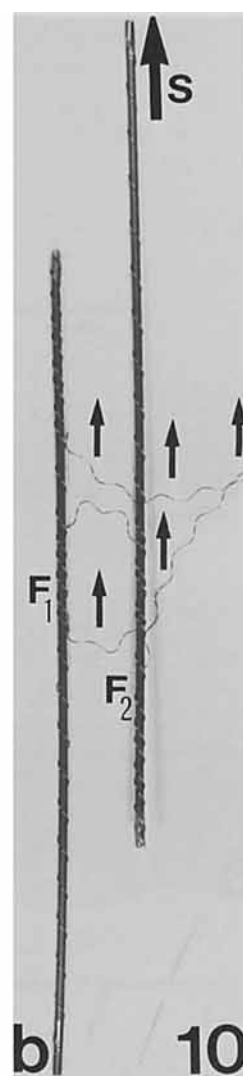
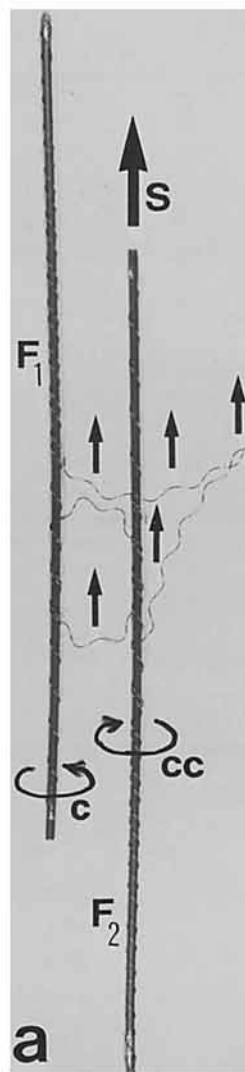
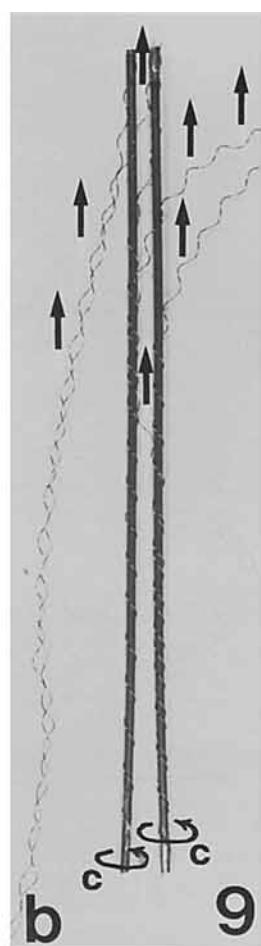
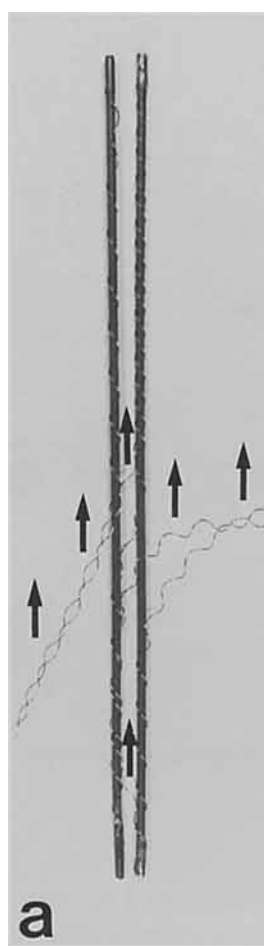
Figs. 6–8. Displacements along cytoskeletal elements.

Fig. 6. A rod-helix complex is rotated (arrow R) by the winding of an electromotor-driven helix (arrow r). The helix moves a particle (long arrow), produces helical waves (arrows W), and slides into the opposite direction (arrow S).

Fig. 7. The side arms 1, 2, 3, and 5 are wound and the side arms 4 and 6 are unwound by the rotating helix. Particles connected with 1 and 5 are transported (arrows).

Fig. 8. The models of two simplified microtubules contain only five protofilaments each. All rotate in the same direction. They are connected by helical crossbridges that are displaced (arrows) by winding and unwinding and move a particle.





tions and winding of smaller filaments [Jarosch and Foissner, 1982]. Recent AVEC microscopy, however, revealed bidirectional particle transport along single isolated MTs where the particles could pass and overtake each other [Hayden and Allen, 1984; Allen et al, 1985; Schnapp et al, 1985]. This seems to be only consistent with the conception that a transporting MT is a bundle of rotating protofilaments, as is demonstrated by the model of Figure 8 for simplified MTs that contain five protofilaments each, which rotate in the same direction. The protofilaments are connected and crossbridged by helices that are wound and unwound. The crossbridges are displaced together with a particle as it may also be expected for the many crossbridges between neurotubules and neurofilaments (Figs. 11 and 12). The bidirectional particle transport could be explained if protofilaments of a single MT rotate in opposite directions or are connected with helices of opposite coiling direction. This is consistent with the observation that the particles traveled frequently in the same (forward) direction in which the MTs glided [Allen et al, 1985].

PARALLEL AND ANTIPARALLEL SLIDING

The sliding motion of the rods depends on the mode of helix winding and the direction of the rotations, respectively [Jarosch and Foissner, 1982]. In the model of Figure 9, both rods rotate in the same direction and displace the side arms (arrows) if the resistance of the medium is small. In a medium of high viscosity, they would screw in the opposite direction (parallel sliding). If the two rods rotate in opposite directions (arrow c and arrow cc in Fig. 10), they would screw in opposite directions (antiparallel sliding). However, the resulting motions depend here additionally on the resistance against the sliding motions. If F_1 is fully prevented from sliding

as shown in Figure 10, all waves, side arms, and crossbridges are displaced as in Figure 9 (small arrows), but F_2 overtakes this motion with double velocity (large arrow S). If F_1 slides back, three velocities result, which are related by the formula

$$V_w = (V_{F_2} - V_{F_1})/2,$$

where V_w is the velocity of the helical waves that are unidirectional on both counterrotating elements, V_{F_2} is the sliding velocity of the quicker element, and V_{F_1} is the sliding velocity of the slower element in opposite direction [Jarosch and Foissner, 1983].

The distances between rods that rotate in the same direction (Fig. 9) are smaller than those that rotate in opposite directions (Fig. 10). This is because the helices unwind and wind directly from one rod to the other when they rotate in the same direction but bend back between counterrotating rods. This seems to be responsible for the different distances between dendritic (Fig. 11) and axonic neurofilaments and MTs (Fig. 12) as described by Hirokawa, Glicksman, and Willard [1984] and Hirokawa, Bloom, and Vallee [1985]. The crossbridges are indeed different in the two cases. This interpretation presupposes for the native state (see above) that all protofilaments of a single MT rotate in the same direction.

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Figs. 9–12. Possible behavior of neurotubules and neurofilaments.

Fig. 9. Displacement of crossbridges and side arms (arrows) along rods that rotate in the same direction.

Fig. 10. Displacement of crossbridges and side arms (arrows) along counterrotating rods (arrows c and cc). The sliding motion of F_1 is fully prevented. Therefore, F_2 slides (arrow S) with the double velocity of the crossbridges.

Fig. 11. Dendritic neurofilaments and microtubules that show crossbridging as in Figure 9 are closely distanced [from Hirokawa et al, 1984, with permission].

Fig. 12. Axonal microtubules and neurofilaments that show crossbridging similar to that in Figure 10 are spaced farther apart than dendritic ones [compare Fig. 11, from Hirokawa et al, 1984, 1985, with permission].

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